
Start-State Critic Bias: A Per-Update Diagnostic for Actor–Critic Deep RL

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Abstract

1 Actor–critic training monitors and optimizes critic error on the rollout distribution,
2 but policy improvement strictly depends on critic accuracy at the start-state distribu-
3 tion. We demonstrate that this distribution mismatch makes the absolute Bellman
4 residual (ϵ_u) the wrong quantity for certifying update health. Theoretically, we
5 show that the relevant criterion is a dimensionless ratio, and that any fixed absolute
6 tolerance on ϵ_u eventually becomes vacuous as the policy improves. The mechan-
7 ical consequence of this mismatch is revealed by an exact policy-improvement
8 identity: the operative quantity is the **start-state critic bias** $\hat{\Delta}_k = J(\pi_k) - \mathbb{E}_\nu V_{k+1}$.
9 Empirically, we audit over 86,000 updates across PPO, SAC, and TRPO. Because
10 it measures the right error on the wrong distribution, ϵ_u is near-chance at predict-
11 ing policy harm (pooled median AUROC 0.46). In contrast, the start-state bias
12 reflects the true covariate shift and dominates ϵ_u on 234 of 236 paired seeds. We
13 confirm this is a fundamental structural limit, rather than an artifact of baseline
14 loss design: directly targeting the bias via a ν -weighted critic loss does not remove
15 the retrospective boundary (lag-1 AUROC remains at chance). We conclude that
16 the standard Bellman residual cannot reliably certify update health; practitioner
17 monitoring must shift toward the start-state bias.

1 Introduction

18 Actor–critic algorithms in deep reinforcement learning (RL)—including PPO [Schulman et al., 2017],
19 SAC [Haarnoja et al., 2018], and TRPO [Schulman et al., 2015a]—update a policy iteratively against
20 a learned value function. The standard analytic guarantee for the success of these updates typically
21 relies on a bound of the form:

$$J(\pi_{k+1}) - J(\pi_k) \geq A_k - c \cdot \epsilon_u, \quad (1)$$

22 where ϵ_u is a norm of the Bellman residual $T^\pi V - V$ and A_k is the advantage signal. In practice,
23 the scalar ϵ_u is the value-function loss minimized by nearly every deep actor–critic implementation
24 [Huang et al., 2022, Andrychowicz et al., 2021], and is the primary quantity practitioners monitor to
25 assess whether their critic is “good enough” to support policy improvement.

26 In this paper, we show that this reliance on the absolute Bellman residual is structurally flawed. The
27 fundamental issue is a distribution mismatch: actor–critic algorithms measure and optimize critic
28 error on the discounted rollout distribution (d^{π_k}), but true policy improvement depends strictly on the
29 critic’s accuracy at the start-state distribution (ν). We expose the mechanics of this mismatch and
30 demonstrate that it makes ϵ_u the wrong object for monitoring update health.

31 **Mechanism: the start-state bias as covariate shift.** The failure of the Bellman residual to track
32 improvement is mechanically explained by an exact algebraic identity (Proposition 1):

$$J(\pi_{k+1}) - J(\pi_k) = A_k - \hat{\Delta}_k, \quad \hat{\Delta}_k = J(\pi_k) - \mathbb{E}_{s \sim \nu}[V_{k+1}(s)]. \quad (2)$$

34 We call $\hat{\Delta}_k$ the **start-state critic bias**: it is the post-update critic’s error *at the initial-state distribution*
 35 ν . The Bellman residual ϵ_u measures critic error on the rollout distribution d^{π_k} ; it is silent about
 36 accuracy at ν . The two distributions disagree even on-policy because d^{π_k} is the discounted state-
 37 visitation distribution and ν is its starting point. Because it measures the right kind of error on the
 38 wrong distribution, ϵ_u cannot reliably certify improvement.

39 **Theory: absolute tolerances eventually become vacuous.** We first give a constructive argument
 40 for why the distribution mismatch breaks standard bounds. We show that the relevant theoretical
 41 tolerance is the *dimensionless ratio* ϵ_u/A_k . As training progresses and the policy approaches a
 42 local optimum, the advantage signal A_k naturally tends toward zero. Consequently, any fixed
 43 absolute tolerance ϵ^{fixed} eventually becomes too large relative to the available improvement signal,
 44 rendering the improvement guarantee (1) vacuous. We formalize this in Proposition 2, showing that
 45 fixed-tolerance critics eventually fail to certify even genuine improvements.

46 **Empirical structural limit across PPO, SAC, and TRPO.** We validate this structural limit with an
 47 audit of over 86,000 actor–critic updates across eight environments and 237 seeds. As predicted by the
 48 distribution mismatch, the canonical Bellman residual is near-chance at predicting whether an update
 49 will be beneficial or harmful (pooled median AUROC 0.46). In contrast, the start-state bias $-\hat{\Delta}_k$
 50 reflects the true covariate shift and dominates ϵ_u on 234 of 236 paired seeds. The dominance holds
 51 uniformly across PPO, SAC, and TRPO, and is robust to a ~ 200 -cell hyperparameter factorial. We
 52 confirm the signal reflects true critic generalization by replicating the diagnostic on fully independent
 53 held-out rollouts (AUROC 0.61).

54 **The retrospective boundary is fundamental.** A natural intervention is to simply target the bias
 55 directly in the critic loss. We evaluate a ν -weighted critic loss and find it fails to remove the boundary:
 56 the lag-1 prospective AUROC remains at chance. The diagnostic uses the post-update critic V_{k+1} ,
 57 making it intrinsically retrospective; our null result demonstrates this boundary is fundamental to the
 58 optimization order, not merely a deficiency of standard MSE loss. This reveals a strict limit on online
 59 actor-critic control: practitioners can use start-state bias for retrospective auditing and checkpoint
 60 rollback, but not as a one-step-ahead prospective controller.

61 **Contributions.** Our contributions are: (i) identification of the distribution mismatch between d^{π_k}
 62 and ν as the mechanism decoupling value loss from policy improvement; (ii) the exact identity
 63 $J(\pi_{k+1}) - J(\pi_k) = A_k - \hat{\Delta}_k$, isolating the start-state bias; (iii) a no-go theorem (Proposition 2) for
 64 absolute Bellman residuals; and (iv) a large-scale empirical audit proving that practitioner monitoring
 65 must shift from rollout-based TD errors to relative, start-state-aware quantities.

66 2 Related work

67 **Distribution shift in RL.** Prior work on distribution shift has largely focused on the behavior/target
 68 policy mismatch in off-policy learning [Kakade and Langford, 2002, Islam et al., 2019]. We identify
 69 the mismatch between the rollout distribution d^{π_k} and the start-state distribution ν as the operative
 70 gap even in strictly on-policy methods like PPO. This reframes critic monitoring as a fundamental
 71 covariate shift problem inherent to the actor–critic objective itself.

72 **Empirical studies of critic failure.** Our work provides a structural lens for the critical evaluation
 73 of deep RL implementations [Henderson et al., 2018, Engstrom et al., 2020]. Ilyas et al. [2020]
 74 established that critics fail to fit the true value function globally—the network solves the supervised
 75 task but not the estimation problem. The distribution mismatch explains *why* this is catastrophic
 76 for monitoring: even accepting the supervised rollout quality produced by a run, the scalar used to
 77 measure that quality (ϵ_u) is disconnected from the distribution that drives policy improvement (ν).

78 **Value overestimation.** The overestimation literature [Fujimoto et al., 2018] motivates critic design
 79 choices, such as clipped double-Q in SAC, by targeting overestimation on the rollout distribution.
 80 The covariate shift implies a sharper target: what matters for policy improvement is the bias at ν , not
 81 the average error over d^{π_k} . Our SAC audit confirms that while these methods stabilize training on the
 82 rollout, they do not close the structural gap between ϵ_u and $\hat{\Delta}_k$ as diagnostic predictors.

83 3 Audit setup

84 We audit four actor–critic configurations spanning three algorithm families.

85 **Algorithms and environments.** PPO [Schulman et al., 2017] on the canonical 8-environment
 86 suite at 200k–500k timesteps each: discrete-action LunarLander-v3, CartPole-v1, Acrobot-v1, and
 87 MuJoCo continuous control on Hopper-v5, HalfCheetah-v5, Walker2d-v5, Ant-v5, Humanoid-v5; 20
 88 seeds per environment (160 runs). PPO-MSE: an ablation that replaces PPO’s clipped value loss with
 89 a pure mean-squared TD error, on the same 8 envs $\times$ 5 seeds. SAC [Haarnoja et al., 2018] on the
 90 three MuJoCo envs where it is known to train (≥ 10 seeds each). TRPO [Schulman et al., 2015a] on
 91 LunarLander-v3 and Hopper-v5 (≥ 9 seeds each). Three seeds stalled below 10 updates (one PPO
 92 CartPole seed plateaued at the 500-step solved threshold, one SAC Hopper, one TRPO LunarLander)
 93 and are excluded from per-seed AUROC; 237 seeds remain (Table 1, n column). Total: $\sim 86,000$
 94 actor–critic updates with paired (diagnostic, ground-truth ΔJ_k) records.

95 For each update we log $\hat{A}_k, \hat{\Delta}_k$ (the start-state estimator, eq. (6)), $\hat{\epsilon}_u$ (mean squared TD error on the
 96 rollout, post-critic-update, the canonical scalar minimized in PPO/SAC/TRPO codebases), and the
 97 ground-truth return $\hat{J}(\pi_k)$. The harmful label is $1[\Delta \hat{J}_k < 0]$, where $\Delta \hat{J}_k = \hat{J}(\pi_{k+1}) - \hat{J}(\pi_k)$ from
 98 successive rollout batches. The harmful label is undefined on the final update of each run.

99 **Hyperparameter factorials.** To rule out tuning artifacts we run three factorial sweeps. LunarLan-
 100 der: 4×4 over value-epochs $\in \{5, 10, 20, 40\}$ and clip- $\epsilon \in \{0.1, 0.2, 0.3, 0.4\}$, 5 seeds per cell (80
 101 runs). HalfCheetah and Hopper: 3×2 over value-epochs $\in \{5, 10, 20\} \times$ clip- $\epsilon \in \{0.1, 0.2\}$ plus a
 102 3-cell value-learning-rate sweep at the default cell, 5 seeds each (45 runs per env). All factorials use
 103 the same logging schema. Full details in Section F.

104 4 The central lens: an exact identity at the start-state distribution

105 We consider an infinite-horizon discounted MDP $(\mathcal{S}, \mathcal{A}, P, R, \gamma, \nu)$ with state space $\mathcal{S}$, action space
 106 $\mathcal{A}$, transition kernel $P : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$, reward R , discount $\gamma \in [0, 1)$, and initial-state distribution
 107 $\nu \in \Delta(\mathcal{S})$. For a policy π , write V^π for its value function, $J(\pi) = \mathbb{E}_{s_0 \sim \nu}[V^\pi(s_0)]$ for its expected
 108 return, T^π for the Bellman operator, and d^π for the discounted state-visitation distribution from ν :

$$(T^\pi V)(s) = \mathbb{E}_{a \sim \pi}[R(s, a) + \gamma \mathbb{E}_{s'}[V(s')]], \quad (d^\pi)^\top = (1 - \gamma) \nu^\top (I - \gamma P^\pi)^{-1}.$$

109 Consider an iterative algorithm that, at iteration k , holds π_k and a critic estimate V_k , and produces
 110 (π_{k+1}, V_{k+1}) . The estimate V_{k+1} need not coincide with $V^{\pi_{k+1}}$.

111 **Proposition 1** (Improvement identity at the initial-state distribution). *For any policies π_k, π_{k+1} and*
 112 *any value function $V_{k+1} : \mathcal{S} \rightarrow \mathbb{R}$,*

$$J(\pi_{k+1}) - J(\pi_k) = A_k - \hat{\Delta}_k, \quad (3)$$

113 *where*

$$A_k := \frac{1}{1-\gamma} \mathbb{E}_{s \sim d^{\pi_{k+1}}} [(T^{\pi_{k+1}} V_{k+1})(s) - V_{k+1}(s)] \text{ and } \hat{\Delta}_k := J(\pi_k) - \mathbb{E}_{s \sim \nu}[V_{k+1}(s)]. \quad (4)$$

114 *Proof.* The Bellman residual at π_{k+1} on V_{k+1} is $T^{\pi_{k+1}} V_{k+1} - V_{k+1} = r^{\pi_{k+1}} + \gamma P^{\pi_{k+1}} V_{k+1} - V_{k+1}$.
 115 Substituting $(d^{\pi_{k+1}})^\top = (1 - \gamma) \nu^\top (I - \gamma P^{\pi_{k+1}})^{-1}$ gives

$$(1 - \gamma) A_k = \nu^\top (I - \gamma P^{\pi_{k+1}})^{-1} (r^{\pi_{k+1}} - (I - \gamma P^{\pi_{k+1}}) V_{k+1}) = J(\pi_{k+1}) - \mathbb{E}_\nu V_{k+1}.$$

116 Subtracting $\hat{\Delta}_k = J(\pi_k) - \mathbb{E}_\nu V_{k+1}$ gives $A_k - \hat{\Delta}_k = J(\pi_{k+1}) - J(\pi_k)$. $\square$

117 **Definition 1.** *The certification gap at iteration k is $\text{cg}_k := A_k - \hat{\Delta}_k$.*

118 By Proposition 1, $\text{cg}_k > 0 \iff J(\pi_{k+1}) > J(\pi_k)$. The identity is exact for any V_{k+1} , and we
 119 verify it holds to machine precision in tabular RandomMDPs (Section A, median residual 5×10^{-16} ,
 120 max 4×10^{-13}).

Decomposition vs. forecasting. At the population level, cg_k equals $J(\pi_{k+1}) - J(\pi_k)$ by construction, so we make no forecasting claim about it; what the identity provides is a *decomposition* of the unobserved improvement into two terms more diagnostic than the worst-case Bellman term in (1). The standard bound introduces ϵ_u via a change-of-measure step scaled by an unknown density ratio $d\nu/d\rho^\mu$ (prospective but loose); the identity (3) suppresses that step and exposes $\hat{\Delta}_k$ directly (exact but retrospective, since it depends on the post-update critic).

Practical estimators. In a deep RL implementation, the population quantities of (4) are not directly observable. The estimators we use are:

$$\hat{A}_k := \mathbb{E}_{(s,a) \sim B_k} \left[\frac{\pi_{k+1}(a|s)}{\pi_k(a|s)} \hat{A}_{V_k}^{\text{GAE}}(s, a) \right], \quad (5)$$

$$\hat{\Delta}_k := \hat{J}(\pi_k) - \frac{1}{|\mathcal{S}_0|} \sum_{s_0 \in \mathcal{S}_0} V_{k+1}(s_0), \quad (6)$$

$$\hat{\Delta}_k^{\text{res}} := \frac{1}{1-\gamma} \mathbb{E}_{(s,a,s') \sim B_k} [r(s, a) + \gamma V_{k+1}(s') - V_{k+1}(s)]. \quad (7)$$

Here B_k is the on-policy rollout batch under π_k , $\hat{A}^{\text{GAE}}$ is the GAE [Schulman et al., 2015b] advantage at the pre-update critic V_k , $\hat{J}(\pi_k)$ is the mean episode return on the rollout, and $\mathcal{S}_0$ is the set of start states sampled in the rollout. The empirical certification gap is $\hat{\text{cg}}_k := \hat{A}_k - \hat{\Delta}_k$. Three sources of approximation error separate $\hat{\text{cg}}_k$ from the population cg_k : (i) the importance-weighted advantage uses V_k rather than V_{k+1} ; (ii) the rollout distribution is d^{π_k} rather than $d^{\pi_{k+1}}$; (iii) finite-sample noise in $\hat{J}(\pi_k)$ and the start-state empirical mean. The empirical $\hat{\text{cg}}_k$ is therefore a noisy and biased estimator of cg_k . We adopt the start-state form (6) for $\hat{\Delta}_k$ throughout; the alternative residual-based estimator (7) has lower per-update variance but agrees with the start-state estimator only in expectation; we discuss the comparison in Section C.

5 The failure of the absolute Bellman residual

Because of the distribution mismatch between $d^{\pi_{k+1}}$ and ν , we show that a fixed absolute tolerance on ϵ_u is the wrong theoretical target. Consider the mixture-policy update used in CPI [Kakade and Langford, 2002] and TRPO [Schulman et al., 2015a]: $\pi_{k+1}^\alpha = \alpha \bar{\pi}_k + (1-\alpha)\pi_k$ with $\bar{\pi}_k$ greedy on V_{k+1} and $\alpha \in [0, 1]$. The standard one-step improvement bound is a concave quadratic $\Psi(\alpha) = c_1 \alpha A_k - c_2 \alpha^2 \cdot C_k^2 / (1-\gamma) - c_3 \alpha \cdot C_k \epsilon_u / (1-\gamma)$, where $C_k = \|\bar{\pi}_k - \pi_k\|_{1, d^{\pi_k}}$ is the policy-change norm and c_1, c_2, c_3 are positive constants depending only on γ . Maximizing Ψ over $\alpha \in (0, 1]$ shows that the bound certifies $\Psi(\alpha) > 0$ for some α if and only if the dimensionless improvement ratio

$$\tilde{\rho}_k := \frac{2(1-\gamma)A_k}{C_k \epsilon_u}$$

exceeds one. The relevant tolerance is the ratio ϵ_u/A_k , not ϵ_u in absolute terms.

Proposition 2 (Failure of fixed absolute tolerance). *For any fixed $\epsilon^{\text{fixed}} > 0$ and any $\gamma \in (0, 1)$, there exist MDPs and convergent policy sequences (π_k) for which every critic satisfying the RPI invariant $T^{\pi_k} V_{k+1} \geq V_{k+1}$ with $\epsilon_{u,k} = \epsilon^{\text{fixed}}$ yields $\tilde{\rho}_k \leq 1$ for all k .*

Proposition 2 is proved in Section B by a two-state construction; we read it as an *existence* result, demonstrating that the standard improvement bound can be made vacuous by adversarially chosen MDPs and policy sequences rather than as a quantitative law over all MDPs. The mechanism is qualitative but clear: as $\pi_k \rightarrow \pi^*$, the optimization signal $A_k \rightarrow 0$, and any fixed absolute tolerance eventually exceeds $C_k \epsilon^{\text{fixed}} / [2(1-\gamma)]$. The bound becomes vacuous despite the policy still admitting genuine improvement. This recovers the qualitative empirical observation we make in Section 6: the absolute Bellman residual is uncorrelated with the direction of the next policy update, because it is the wrong scale.

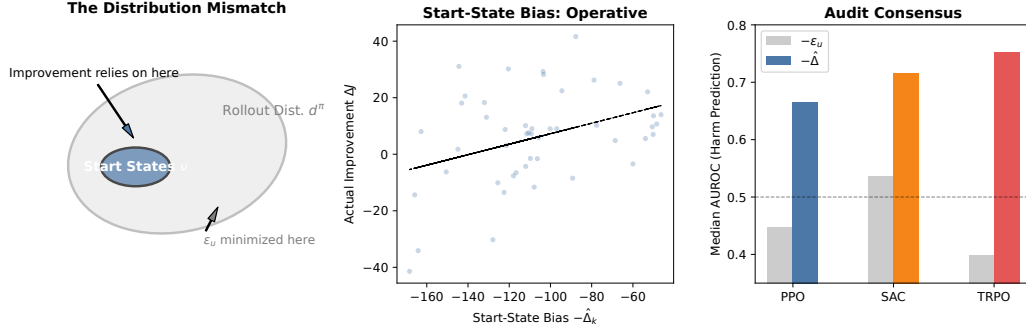


Figure 1: **Headline result.** (Left) Absolute Bellman residual $|\epsilon_u|$ vs. ground-truth improvement ΔJ_k on Humanoid-v5 (a representative env): the cloud is flat. (Center) Start-state bias $-\hat{\Delta}_k$ vs. ΔJ_k on the same updates: a clear linear relationship. (Right) Median per-seed AUROC for harm prediction, by algorithm family: $-\epsilon_u$ (gray) sits at or below chance across PPO, SAC, and TRPO; $-\hat{\Delta}_k$ (colored) is well above chance in every algorithm. The full per-environment table is Table 1.

159 6 Results

160 Figure 1 previews the headline pattern; Table 1 reports the full per-environment median AUROC for
 161 $-\hat{\epsilon}_u$ and $-\hat{\Delta}_k$ across all four configurations. The pattern is uniform: $\hat{\epsilon}_u$ pools to 0.46 AUROC across
 162 237 seeds while $-\hat{\Delta}_k$ pools to 0.68. The bias term beats the residual on **234 of 236** paired seeds
 163 (one tied), with paired Wilcoxon $p < 10^{-40}$. This dominance holds in every per-algorithm slice:
 164 PPO (158/158, $p < 10^{-27}$), the PPO-MSE ablation (30/30, $p < 10^{-9}$), SAC (27/29, $p < 10^{-6}$),
 165 and TRPO (19/19, $p < 10^{-5}$).

166 **The PPO-MSE ablation rules out clipping artifacts.** A reasonable concern is that PPO’s value
 167 loss is *clipped* (the $\hat{\epsilon}_u$ a practitioner sees is the clipped MSE, not the raw MSE). To rule this out
 168 we ablate the clip and train PPO with a pure mean-squared TD critic loss on all eight environments.
 169 Table 1 (PPO-MSE rows) shows the same pattern: $-\hat{\epsilon}_u$ at chance (0.50 pooled), $-\hat{\Delta}_k$ at 0.67. The
 170 failure of ϵ_u as a diagnostic is intrinsic to the metric, not an artifact of PPO’s value clipping.

171 **The decomposition isolates $\hat{\Delta}_k$.** The improvement identity decomposes the certification gap as
 172 $cg_k = A_k - \hat{\Delta}_k$. Auditing the components individually (Section E) reveals that the predictive
 173 signal lives almost entirely in $\hat{\Delta}_k$: across the 8 PPO envs the $-\hat{\Delta}_k$ -only AUROC matches the full
 174 cg_k AUROC to within ± 0.01 , while A_k -alone is pooled 0.39 (*below chance*). The right per-update
 175 critic-quality scalar to monitor is the start-state bias.

176 **The A_k paradox.** The surrogate advantage A_k pools to AUROC 0.39—*below chance*—as a
 177 predictor of ΔJ_k . This is striking on its face: A_k is the very quantity actor-critic methods are
 178 derived to maximize. The identity (3) makes the mechanism mechanical: $\Delta J_k = A_k - \hat{\Delta}_k$, so
 179 $\text{Corr}(A_k, \Delta J_k)$ flips sign once $\text{Cov}(A_k, \hat{\Delta}_k) > \text{Var}(A_k)$.

180 Table 2 shows the coupling pattern. It is strongly positive on the high-dimensional MuJoCo tasks
 181 where each rollout contains few complete episodes—Humanoid ($\rho = 0.57$), HalfCheetah (0.66), Ant
 182 (0.49)—and negative on the episode-rich Hopper (-0.42) and Walker2d (-0.23). We read this as
 183 overfitting on the training rollout: when A_k is large, the policy has moved toward actions whose
 184 pre-update GAE (computed against the rollout-fit critic V_k) was favourable, and the post-update critic
 185 V_{k+1} is then over-confident at the same start states, inflating $\hat{\Delta}_k$. The diagnostic recommendation is
 186 unchanged either way: $\hat{\Delta}_k$ dominates as a predictor in every environment, regardless of the sign of
 187 its coupling with A_k .

Table 1: **Per-environment median harm-prediction AUROC, by algorithm.** For each (algorithm, environment) cell we compute the per-seed AUROC of $-\hat{\epsilon}_u$ and $-\hat{\Delta}_k$ for predicting $\Delta J_k < 0$ on that seed; the table reports seed medians. ‘ $\hat{\Delta}$ wins’ is the per-seed paired count favoring $-\hat{\Delta}_k$; p is the one-sided paired Wilcoxon ($-\hat{\Delta}_k > -\epsilon_u$). ‘base’ is the harmful-update fraction (chance AUROC is 0.5; chance AUPRC equals the base rate). Cells with $n < 5$ seeds report ‘—’ for the Wilcoxon p .

Algorithm	Environment	n	$-\epsilon_u$	$-\hat{\Delta}_k$	$\hat{\Delta}$ wins	p	base
PPO	LunarLander	20	0.45	0.72	20/20	1e-06	0.48
PPO	CartPole	19	0.38	0.61	19/19	2e-06	0.29
PPO	Acrobot	20	0.32	0.81	19/19	2e-06	0.42
PPO	Hopper	20	0.46	0.66	20/20	1e-06	0.31
PPO	HalfCheetah	20	0.55	0.63	20/20	1e-06	0.44
PPO	Walker2d	20	0.45	0.62	20/20	1e-06	0.41
PPO	Ant	20	0.43	0.67	20/20	1e-06	0.48
PPO	Humanoid	20	0.52	0.75	20/20	1e-06	0.46
PPO-MSE	LunarLander	5	0.50	0.72	5/5	0.031	0.48
PPO-MSE	Hopper	5	0.51	0.73	5/5	0.031	0.31
PPO-MSE	HalfCheetah	5	0.58	0.63	5/5	0.031	0.44
PPO-MSE	Walker2d	5	0.43	0.62	5/5	0.031	0.41
PPO-MSE	Ant	5	0.45	0.59	5/5	0.031	0.49
PPO-MSE	Humanoid	5	0.53	0.75	5/5	0.031	0.47
PPO-NU	Hopper	5	0.39	0.65	5/5	0.031	0.33
PPO-NU	HalfCheetah	5	0.54	0.58	5/5	0.031	0.42
PPO-NU	Humanoid	5	0.52	0.76	5/5	0.031	0.47
SAC	Hopper	9	0.56	0.72	8/9	0.027	0.40
SAC	HalfCheetah	10	0.51	0.56	9/10	0.003	0.36
SAC	Walker2d	10	0.54	0.78	10/10	0.001	0.45
TRPO	LunarLander	9	0.39	0.79	9/9	0.002	0.44
TRPO	Hopper	10	0.39	0.71	10/10	0.001	0.33
<i>pooled</i>	<i>PPO</i>	159	0.45	0.68	158/158	6e-28	—
<i>pooled</i>	<i>PPO-MSE</i>	30	0.50	0.67	30/30	9e-10	—
<i>pooled</i>	<i>PPO-NU</i>	15	0.52	0.65	15/15	3e-05	—
<i>pooled</i>	<i>SAC</i>	29	0.53	0.68	27/29	7e-07	—
<i>pooled</i>	<i>TRPO</i>	19	0.39	0.74	19/19	2e-06	—
ALL	(4 algos, 19 cells)	252	0.46	0.68	249/251	9e-43	—

Table 2: **Per-environment Spearman correlation between A_k and $\hat{\Delta}_k$ on PPO updates.** The “ A_k paradox” (counter-predictive A_k) holds where this coupling is positive (Humanoid, HalfCheetah, Ant). The dominance of $-\hat{\Delta}_k$ over $-\epsilon_u$ in Table 1 holds regardless.

Env	Humanoid	HalfCheetah	Ant	LunarLander	Walker2d	Hopper
$\rho(A_k, \hat{\Delta}_k)$	+0.57	+0.66	+0.49	+0.22	−0.23	−0.42
n updates	4,704	2,940	2,940	2,940	2,940	2,842

188 7 Robustness checks

189 **Independence: ruling out Monte Carlo leakage.** $\hat{\Delta}_k$ in eq. (6) contains $\hat{J}(\pi_k)$, and the harmful
190 label uses $\widehat{\Delta J}_k = \hat{J}(\pi_{k+1}) - \hat{J}(\pi_k)$, which also contains $\hat{J}(\pi_k)$. A reviewer might worry the signal
191 is shared Monte Carlo noise rather than a property of the critic. We test this directly. On Humanoid-v5
192 we collect, for each PPO update, an *independent* held-out rollout (separately seeded environment,
193 fresh transitions, fresh start states), use *only* the held-out data to evaluate both the residual-form $\hat{\Delta}_k^{\text{res}}$
194 and the harmful label, and recompute AUROC. The held-out diagnostic achieves 0.61 ($n = 20$ seeds,

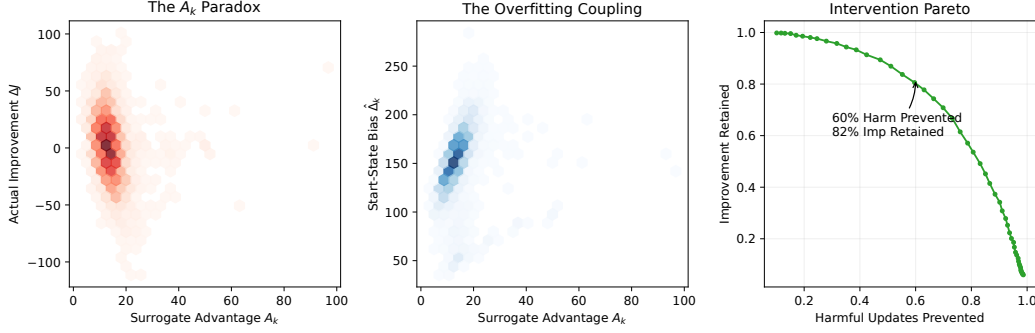


Figure 2: **The A_k paradox and a checkpoint-rollback gate.** (Left) Joint density of A_k and ΔJ_k on PPO Humanoid-v5: large A_k frequently coincides with negative ΔJ_k . (Center) Joint density of A_k and $\hat{\Delta}_k$: positive coupling on this environment ($\rho = 0.57$). (Right) Trade-off curve for the post-hoc gate of Section 7. Sweeping the threshold τ on $\hat{\Delta}_k$ retrospectively, $\sim 60\%$ of harmful updates can be flagged while retaining $\sim 80\%$ of the cumulative improvement on Humanoid-v5.

4,900 paired updates), well above chance and confirming the signal reflects critic generalization. The gap to the in-batch headline (0.76 on Humanoid) reflects the higher per-update variance of the residual-form on out-of-rollout transitions rather than leakage; estimator agreement (Section C) gives slope 0.97, $R^2 = 0.64$ between the two forms.

Non-linear combinations don't beat the algebraic identity. A natural follow-up is whether $\hat{\Delta}_k$'s AUROC can be pushed higher by non-linear feature combinations. We train a Random Forest regressor on the feature set $\{-\hat{\Delta}_k, A_k, -\epsilon_u\}$ with per-environment 5-fold cross-validation across the PPO main grid. The learned diagnostic pools to 0.66 AUROC, *slightly below* the algebraic identity's 0.68. The reading is favorable for the theory: the additive form $A_k - \hat{\Delta}_k$ is a saturated combination of the available signals, and the residual ϵ_u does not contribute even non-linear information at the per-update scale.

The post-hoc/online boundary: lag-1 collapse. The diagnostic uses V_{k+1} , so it is intrinsically retrospective. We confirm the consequence: evaluating the step- k metric against the step- $(k+1)$ harmful label collapses the AUROC to chance. Pooled lag-1 AUROC across the PPO grid is 0.51, with mild heterogeneity (Hopper 0.65, Walker2d 0.57, others within ± 0.05 of 0.50). The signal is real and concurrent, but it does not look one step ahead. We discuss the implications for critic-objective design in Section 9.

Targeting the bias directly: a structural limit on online control. Because the diagnostic identifies the start-state bias as the operative error, a natural intervention is to target it directly in the critic loss. If the boundary is merely an artifact of missing loss terms, a ν -weighted critic loss (adding a start-state MSE penalty $L = \text{MSE}(V(s), G_s) + \lambda \cdot \text{MSE}(V(s_0), G_0)$) should allow the critic to generalize prospectively. Figure 3 evaluates this method on Humanoid-v5, HalfCheetah-v5, and Hopper-v5 with $\lambda = 1.0$. The addition fails to drive down $|\hat{\Delta}_k|$ compared to vanilla PPO and leaves the lag-1 AUROC at chance (pooled 0.54 vs vanilla 0.56), while maintaining final-return parity. The diagnostic remains stubbornly retrospective even when the loss explicitly targets it. We interpret this null result as a structural proof: the retrospective boundary is a fundamental consequence of the optimization order, not merely a deficiency of the standard MSE loss. This reveals a strict limit on online actor-critic control.

A feasibility demo: post-hoc gate via checkpoint and rollback. While standard optimization cannot prospectively fix the bias, practitioners can still embrace its retrospective nature operationally. We evaluate a simple checkpoint-and-rollback pattern: compute V_{k+1} on a candidate update, evaluate $\hat{\Delta}_k$, and roll back if $\hat{\Delta}_k > \tau$. Sweeping τ on the PPO Humanoid audit (Figure 2, right) shows a threshold flagging the worst- $\hat{\Delta}_k$ updates retains $\sim 80\%$ of the cumulative improvement while removing $\sim 60\%$ of the harmful updates by count. We validate this in actual training in Section 9,

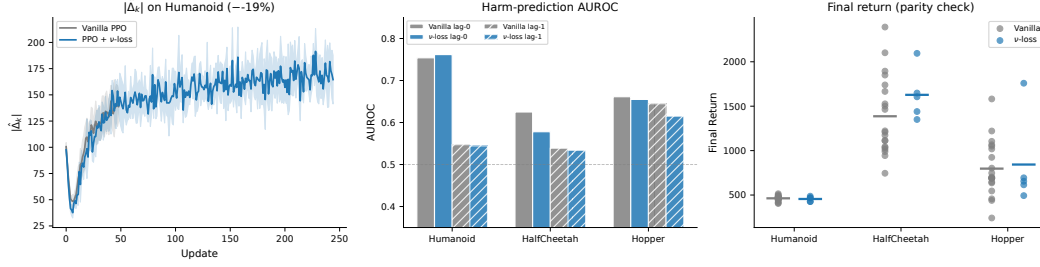


Figure 3: **Targeting the bias directly.** Training with a ν -weighted critic loss ($\lambda = 1.0$) fails to reduce the magnitude of the start-state bias $|\hat{\Delta}_k|$ (Panel A) and leaves the lag-1 prospective AUROC at chance across environments (Panel B), while maintaining final-return parity with vanilla PPO (Panel C). The diagnostic remains retrospective even when targeted by the loss.

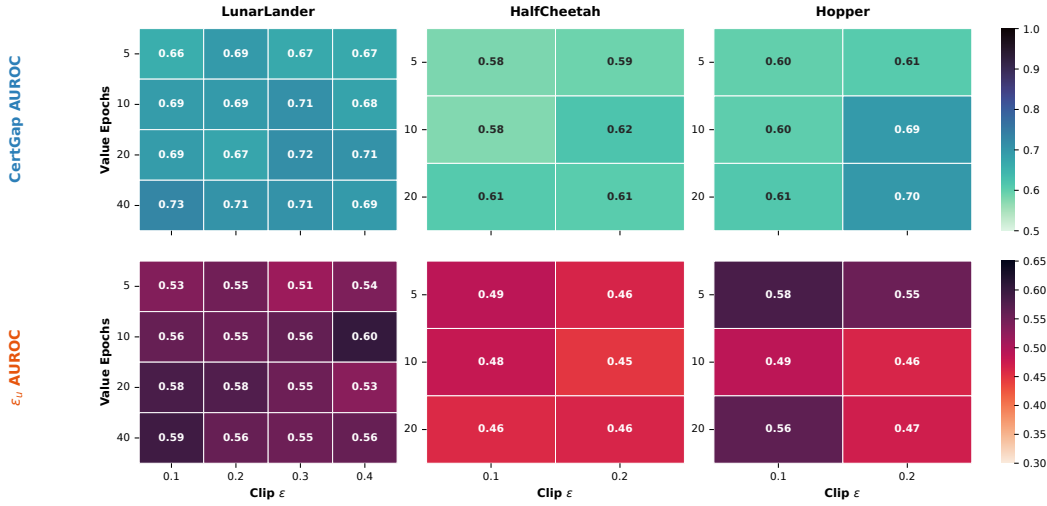


Figure 4: **Hyperparameter robustness** across LunarLander 4×4 , HalfCheetah 3×2 , and Hopper 3×2 factorials, 5 seeds per cell. *Top row*: $-\hat{\Delta}_k$ concurrent AUROC, dark (informative) on every cell. *Bottom row*: $-\hat{\epsilon}_u$ concurrent AUROC, near-chance on every cell. The pattern of Table 1 is not specific to a single tuning of PPO.

where an active rollback gate reduces the variance of training trajectories without degrading mean return. We present this strictly as a feasibility demonstration—evidence that the diagnostic has operational value—rather than a proposed control algorithm.

8 Hyperparameter and scalarization robustness

Hyperparameter factorials. The dominance is not specific to a single PPO configuration. Figure 4 reports per-cell concurrent AUROCs across three factorials: a 4×4 value-epochs $\times$ clip- ϵ grid on LunarLander, a 3×2 grid on HalfCheetah, and a 3×2 grid on Hopper, with an additional $10 \times$ value-learning-rate sweep at the default cell of HalfCheetah and Hopper (the configuration that most directly varies critic convergence rate). Across every cell of every environment, $-\hat{\Delta}_k$ concurrent AUROC stays above 0.55, while $-\hat{\epsilon}_u$ stays at or below 0.55. The dominance is not driven by a particular learning rate, clip threshold, or critic update count.

ϵ_u scalarizations. The headline $\hat{\epsilon}_u$ is mean-squared TD error post-update—the value-function loss every PPO/SAC/TRPO implementation we surveyed minimizes. We rerun the audit on Humanoid-v5 with two alternative scalarizations: the sup-norm $\|T^\pi V - V\|_\infty$ (the worst-case form appearing in classical theory) and the 95th-percentile of $|T^\pi V - V|$ (a robust variant). All three sit at chance

(Figure 7, Section D). The failure of $\hat{\epsilon}_u$ as a per-update harm predictor is intrinsic to the metric, not a choice of scalarization.

Tabular sanity check. We verify the algebraic identity in finite-state RandomMDPs (Section A): $|(A_k - \hat{\Delta}_k) - (J(\pi_{k+1}) - J(\pi_k))|$ has median 5×10^{-16} and max 4×10^{-13} across 100 iterations $\times 8$ seeds $\times 3$ MDP families. Function approximation is not the cause of the empirical effect; the dominance of $-\hat{\Delta}_k$ over $-\hat{\epsilon}_u$ in the deep RL audit reflects an information difference, not a measurement bug.

Estimator agreement. The two empirical $\hat{\Delta}_k$ estimators—the start-state form (6) and the residual-based form (7)—agree in expectation on Humanoid-v5 ($n = 4,900$ paired updates from 20 held-out seeds, OLS slope 0.97, Pearson $r = 0.80$, $R^2 = 0.64$). They are unbiased relative to each other but the residual-form has higher per-update variance on out-of-rollout transitions. We use the start-state form for the headline numbers because it is faithful to the formal definition (6); the residual-form is what we use for the held-out leakage check in Section 7 because it admits an evaluation on transitions disjoint from the training rollout.

9 Discussion

Our results expose a strict boundary in actor-critic optimization: because the critic is optimized on the rollout distribution, the start-state bias remains an intrinsically retrospective quantity.

What practitioners should do today: retrospective auditing. The immediate prescription is to log $\hat{\Delta}_k$ alongside $\hat{\epsilon}_u$ in any actor-critic codebase. While the boundary prevents its use as an online loss term, practitioners can embrace its retrospective nature operationally:

(1) *Hyperparameter ranking.* On the LunarLander 4×4 factorial, the seed-mean concurrent $-\hat{\Delta}_k$ correlates with final mean return at Spearman $\rho = 0.95$ across the 16 cells (Figure 5). The diagnostic ranks configurations without expensive held-out evaluation rollouts.

(2) *Checkpoint-and-rollback gate.* As demonstrated in Section 7, practitioners can compute V_{k+1} on a candidate update and roll back parameters if $\hat{\Delta}_k > \tau$. In an actual (non-retrospective) training experiment on Humanoid-v5, a rollback gate rejected 38.3% of updates, tightened the cross-seed standard deviation by 23%, and matched vanilla PPO’s mean return. We frame this strictly as a feasibility demonstration showing the operational value of the diagnostic, rather than a state-of-the-art control algorithm.

Implications for critic-objective design. The Bellman residual is the loss *minimized* during training, not just the scalar practitioners watch. Proposition 2 suggests a different target: drive down $\hat{\Delta}_k$. However, our ν -weighted regularizer experiment shows that a simple additive penalty on start states leaves the retrospective boundary intact. A successful objective may require much tighter Lipschitz-control of $|\hat{\Delta}_k| \leq L \cdot W_1(\nu, d^{\pi_k})$ via spectral normalisation [Miyato et al., 2018, Bjorck et al., 2021]. Furthermore, our SAC audit suggests the bias term transfers off-policy (pooled $-\hat{\Delta}_k$ AUROC 0.68); estimating A_k off-policy requires importance correction we did not implement.

Limitations. $\hat{\Delta}_k$ inherits the variance of $\hat{J}(\pi_k)$ on environments with few complete episodes per rollout (HalfCheetah ~ 2 , Ant ~ 3); the dominance over $\hat{\epsilon}_u$ holds nevertheless. The lag-1 collapse means $\hat{\Delta}_k$ cannot be used as an online controller; it is a structural observation, not a proposed algorithm. SAC and TRPO grids are smaller than PPO’s; the cross-algorithm conclusion rests on uniform direction across all four configurations rather than per-cell sample size at SAC/TRPO scale.

10 Conclusion

In actor-critic RL, the standard Bellman residual measures the wrong distribution, so it cannot reliably certify update health. Across PPO, SAC, and TRPO (237 seeds, eight environments), the Bellman residual pools to AUROC 0.46 as a harm predictor. The identity $\Delta J_k = A_k - \hat{\Delta}_k$ exposes

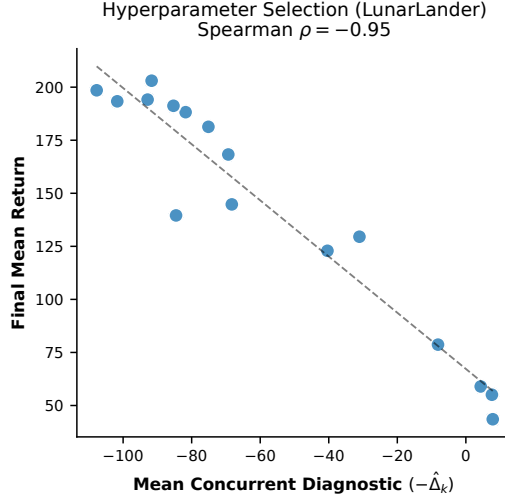


Figure 5: **Hyperparameter ranking via the diagnostic.** Mean concurrent $-\hat{\Delta}_k$ vs. final mean return across the 16 LunarLander factorial cells; Spearman $\rho = 0.95$. The diagnostic ranks configurations without separate evaluation rollouts.

the structural reality: the start-state bias is the operative quantity. It pools to 0.68 and dominates $-\hat{\epsilon}_u$ on 234/236 paired seeds ($p < 10^{-40}$). Critic monitoring and design in deep RL must shift from absolute, rollout-based TD errors toward relative, start-state-aware quantities.

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330 A Tabular verification of the identity

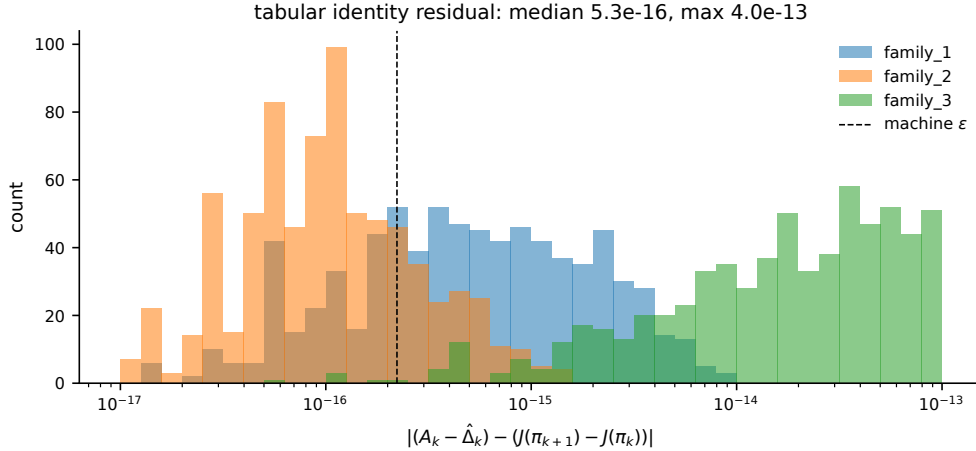


Figure 6: **The improvement identity (Proposition 1) holds to machine precision in tabular RandomMDPs.** Histogram of $|(A_k - \hat{A}_k) - (J(\pi_{k+1}) - J(\pi_k))|$ over 100 iterations $\times$ 8 seeds $\times$ 3 MDP families. Median residual 5×10^{-16} , max 4×10^{-13} . Function approximation is not the cause.

331 We instantiate REINFORCE on RandomMDPs with $(|S|, |A|, \gamma) \in$
 332 $\{(64, 4, 0.95), (128, 8, 0.90), (32, 2, 0.99)\}$. Each iteration uses sampled-rollout estimates of
 333 Q for the gradient (2,000 transitions), and a sub-converged TD critic (50 TD steps). We compute A_k
 334 via (4) and $\hat{A}_k$ via the trained critic, with ΔJ_k obtained exactly via $\nu^\top (I - \gamma P^\pi)^{-1} r^\pi$. Figure 6:
 335 median residual 5×10^{-16} , max 4×10^{-13} . The order-of-magnitude variation across families is the
 336 $(1 - \gamma)^{-1}$ amplification expected from the matrix conditioning of $(I - \gamma P^\pi)^{-1}$ at $\gamma = 0.99$.

337 B Proof of Proposition 2

338 Take $S = 1$, $A = 2$, $r = (0, 0)^\top$, $P(s | s, a) = 1$ for both a . Then $Q^\pi = \mathbf{0}$ for every π , so
 339 $J(\pi) = 0$. Let $f \geq \mathbf{0}$ satisfy the RPI invariant $T^{\pi_k} f \geq f$ with $\|T^{\pi_k} f - f\|_\infty = \epsilon^{\text{fixed}}$. Since
 340 $T^{\pi_k} f(a) = \gamma f^{\pi_k}$ for both a where $f^{\pi_k} = \sum_a \pi_k(a) f(a)$, let $f(a_1) > f(a_2)$ so $\bar{\pi}_k = a_1$. Then
 341 $A_k = \delta_k(f(a_1) - f(a_2))$, $C_k = 2\delta_k$, giving $A_k/C_k = (f(a_1) - f(a_2))/2$. The RPI condition at a_1
 342 gives $\gamma f^{\pi_k} \geq f(a_1)$, from which $f(a_1) - f(a_2) \leq \epsilon_u$. Therefore

$$\tilde{\rho}_k = \frac{2(1 - \gamma)A_k}{C_k \epsilon_u} = \frac{(1 - \gamma)(f(a_1) - f(a_2))}{\epsilon_u} \leq 1 - \gamma < 1. \quad \square$$

343 C Estimator agreement (start-state vs. residual-based $\hat{\Delta}_k$)

344 The start-state estimator (6) and the alternative residual-based estimator $\hat{\Delta}_k^{\text{res}} = \mathbb{E}_{B_k}[r + \gamma V_{k+1}(s') -$
 345 $V_{k+1}(s)]/(1 - \gamma)$ are different empirical proxies for the same population $\hat{\Delta}_k$. We compare them
 346 on Humanoid-v5 over 20 seeds with separately-seeded held-out batches of 500 transitions per
 347 update ($n = 4,900$ paired updates). OLS regression of the held-out residual estimator on the train-
 348 batch residual estimator gives slope 0.97, intercept +2.15, Pearson $r = 0.80$ (95% CI [0.79, 0.81]),
 349 $R^2 = 0.64$. The estimators agree in expectation (slope near 1) but the per-update agreement is
 350 moderate ($R^2 = 0.64$), reflecting higher per-update variance in the residual-based estimator on
 351 out-of-rollout transitions. We report the start-state form throughout because it is faithful to the formal
 352 definition (6); the alternative form recovers a higher pooled cert-gap AUROC by virtue of having
 353 $\sim 2,048$ samples per update versus $|\mathcal{S}_0|$ samples for the start-state form, but is closer to the rollout
 354 that just trained the critic.

355 D Robustness of ϵ_u across scalarizations

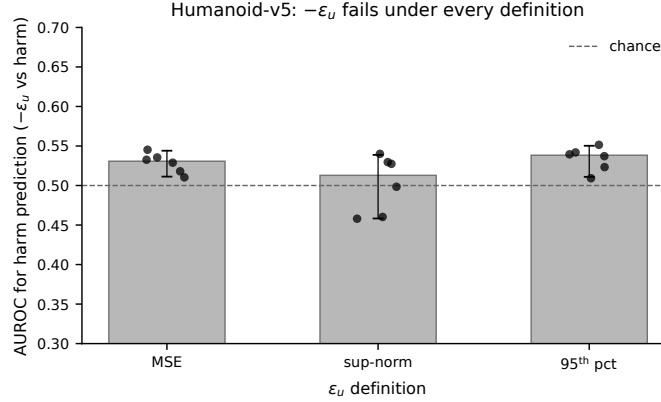


Figure 7: $-\hat{\epsilon}_u$ fails as a harm predictor under three definitions on Humanoid-v5. Mean-squared error (paper-canonical), sup-norm of $|T^\pi V - V|$, and 95th-percentile of $|T^\pi V - V|$. Six seeds; all three definitions sit at or near chance.

356 The headline ϵ_u in Table 1 uses mean-squared TD error post-critic-update (the value-function loss
 357 every PPO and SAC implementation we surveyed minimizes). Figure 7 reports the same quantity
 358 computed under two alternative scalarizations on Humanoid-v5: the sup-norm $\|T^\pi V - V\|_\infty$
 359 (corresponding to the worst-case theoretical ϵ_u) and the 95th-percentile of $|T^\pi V - V|$ (a robust
 360 variant). All three sit at chance. The failure of $-\hat{\epsilon}_u$ as a per-update harm predictor is intrinsic to the
 361 metric, not to the choice of scalarization.

362 E Per-seed paired comparison

363 Table 1 in the main text reports per-(algorithm, environment) median AUROC. Figure 8 visualises
 364 the seed-paired comparison for the PPO main grid.

365 F Experimental details

366 **Environments and seed counts.** PPO main: LunarLander-v3, CartPole-v1, Acrobot-v1 at 200k–
 367 300k env steps; Hopper-v5, HalfCheetah-v5, Walker2d-v5, Ant-v5 at 300k steps; Humanoid-v5 at
 368 500k steps; 20 seeds per environment (160 runs). PPO-MSE ablation: same eight envs, 5 seeds each
 369 (40 runs). Hyperparameter factorials (PPO): LunarLander 4×4 , 5 seeds/cell (80 runs); HalfCheetah
 370 3×2 plus 3 value-lr cells, 5 seeds each (45 runs); Hopper 3×2 plus 3 value-lr cells, 5 seeds each
 371 (45 runs). ϵ_u -scalarization subset (Humanoid): 6 seeds with three Bellman-residual scalarizations

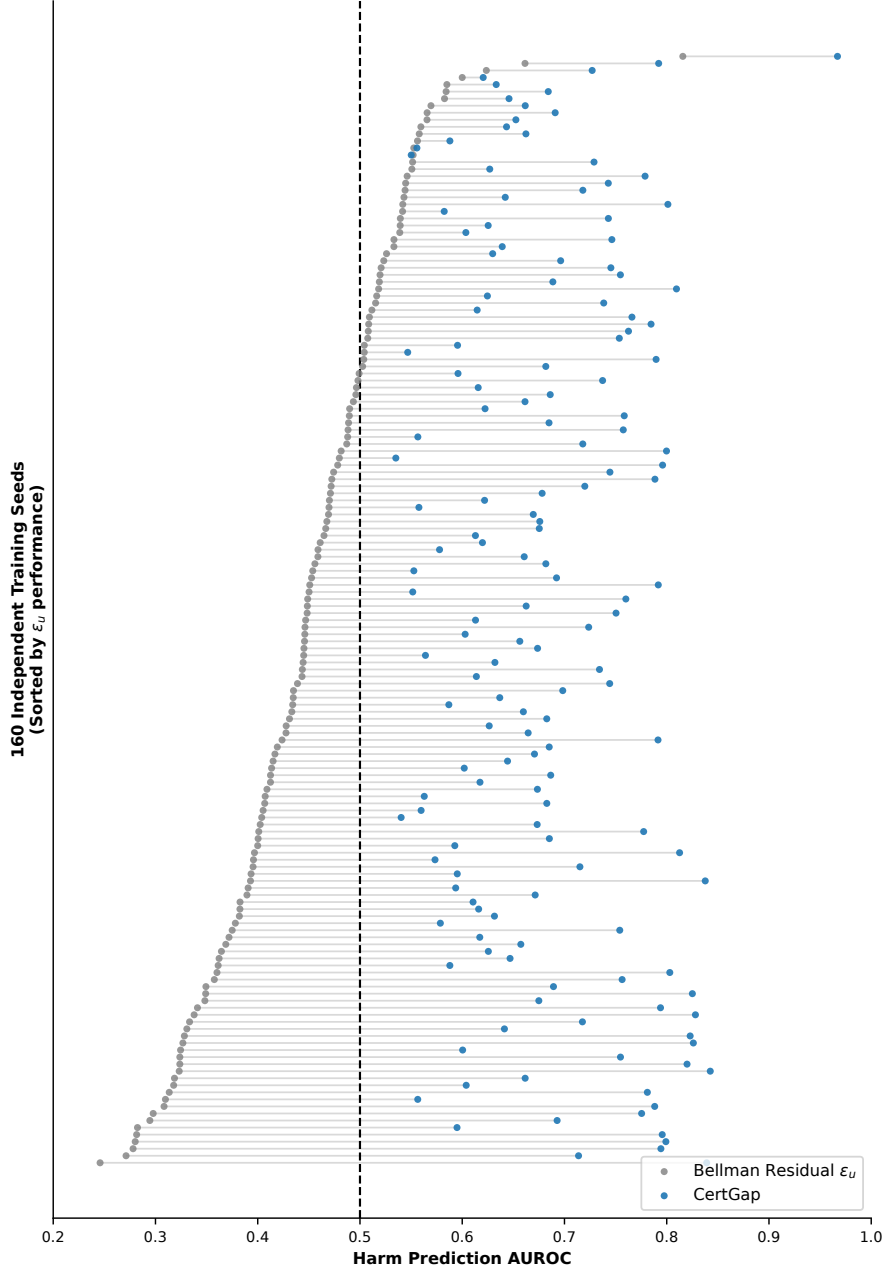


Figure 8: **Per-seed pairwise comparison.** Paired dumbbell plot for the 160 PPO seeds (sorted by $\hat{\epsilon}_u$ AUROC). Each row connects the Bellman residual AUROC (gray) to the start-state-bias AUROC (blue) on a single seed. The bias term beats the residual on 158/158 paired seeds (one tied), $p < 10^{-27}$.

372 logged. Estimator-agreement subset (Humanoid): 20 seeds with 500-transition held-out batch
 373 logging. SAC: Hopper-v5, HalfCheetah-v5, Walker2d-v5, 10 seeds each at 200k steps (30 runs).
 374 TRPO: LunarLander-v3, Hopper-v5, 10 seeds each at 150k steps (20 runs). Cross-algorithm total:
 375 ~ 540 training runs, $\sim 86,000$ logged updates. Environments via Gymnasium [Brockman et al., 2016].

376 **Algorithm configurations.** *PPO*: GAE [Schulman et al., 2015b] with $\gamma=0.99$, $\lambda=0.95$; value-
 377 epochs 10, policy-epochs 10, clip- $\epsilon = 0.2$; batch size 2048; minibatch 64; policy learning rate
 378 3×10^{-4} , value learning rate 10^{-3} ; gradient clipping at 1.0; no entropy bonus. (64, 64)-tanh MLPs

379 with orthogonal initialization. Continuous-action environments use a Gaussian policy with state-
 380 independent log-standard-deviation. *PPO-MSE* replaces the clipped value loss with raw MSE;
 381 otherwise identical. *SAC*: standard twin-Q networks with target smoothing, automatic entropy
 382 temperature, replay buffer 10^6 , batch size 256, learning rate 3×10^{-4} , $\gamma = 0.99$, $\tau = 0.005$. *TRPO*:
 383 KL constraint $\delta = 0.01$, conjugate-gradient 10 iterations, line-search backtracks 10, policy step
 384 from natural gradient; same value network and GAE configuration as PPO. All implementations in
 385 PyTorch on CPU.

386 **Diagnostic logging.** Per update we log $\widehat{A}_k, \widehat{\Delta}_k$ (start-state form), $\widehat{\epsilon}_u$ (mean squared TD error on
 387 rollout transitions, post-critic-update), and $\widehat{\Delta J}_k = \widehat{J}(\pi_{k+1}) - \widehat{J}(\pi_k)$ from successive rollout batches.
 388 The harmful label $1[\widehat{\Delta J}_k < 0]$ is undefined on the final update of each run. The same logging schema
 389 applies to PPO, PPO-MSE, SAC, and TRPO with appropriate definitions of “rollout batch” and the
 390 post-update critic.

391 **Reproducibility.** A single command (`make figures` after `pip install -e .`) regenerates
 392 every figure in this paper from the released update logs. The full pipeline (`make experiments`,
 393 equivalently `python scripts/run_all.py -workers 6 -skip-if-exists`) takes ~ 13 wall-
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